# Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

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For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

## 0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I will use functions, if statements, loops, lists, dataclasses, and user input from Python and Drafter. I plan to use a State dataclass to remember the trip information and a second dataclass for each activity. A loop will go through the activities and add their costs. I want to get better at writing routes, passing information between pages, and testing my functions so I understand how the whole website works.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part will be deciding what information users should enter and designing a simple page for their trip. I already know I want to show the destination, budget, activities, and expenses. The hardest part will probably be making all the routes work together and updating the total whenever someone adds an activity or changes an expense. I will definitely need to make sure the website can handle incorrect inputs without crashing.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think creating and testing the routes will take the most time because each page needs to show the correct information. I will probably start by making the State and Activity dataclasses, then build the trip setup page where the users can enter their destination and budget. After that, I will add the activity list, expense inputs, and then finally the summary with budget calculations. I will test each part before moving on to the next to make sure it makes sense and works.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Other class assignments, my phone, and trying to add too many features could distract me. I will work in shorter time frames spaced out, put my phone away, and stick to the simple budget-focused idea I chose in last week's assignment. I will use my class notes, previous BlockPy assignments, the Drafter documentation, and help from my TA or professor when possibly needed. One question I still have is the best way to test routes that change State, so I will check the class examples before coding.

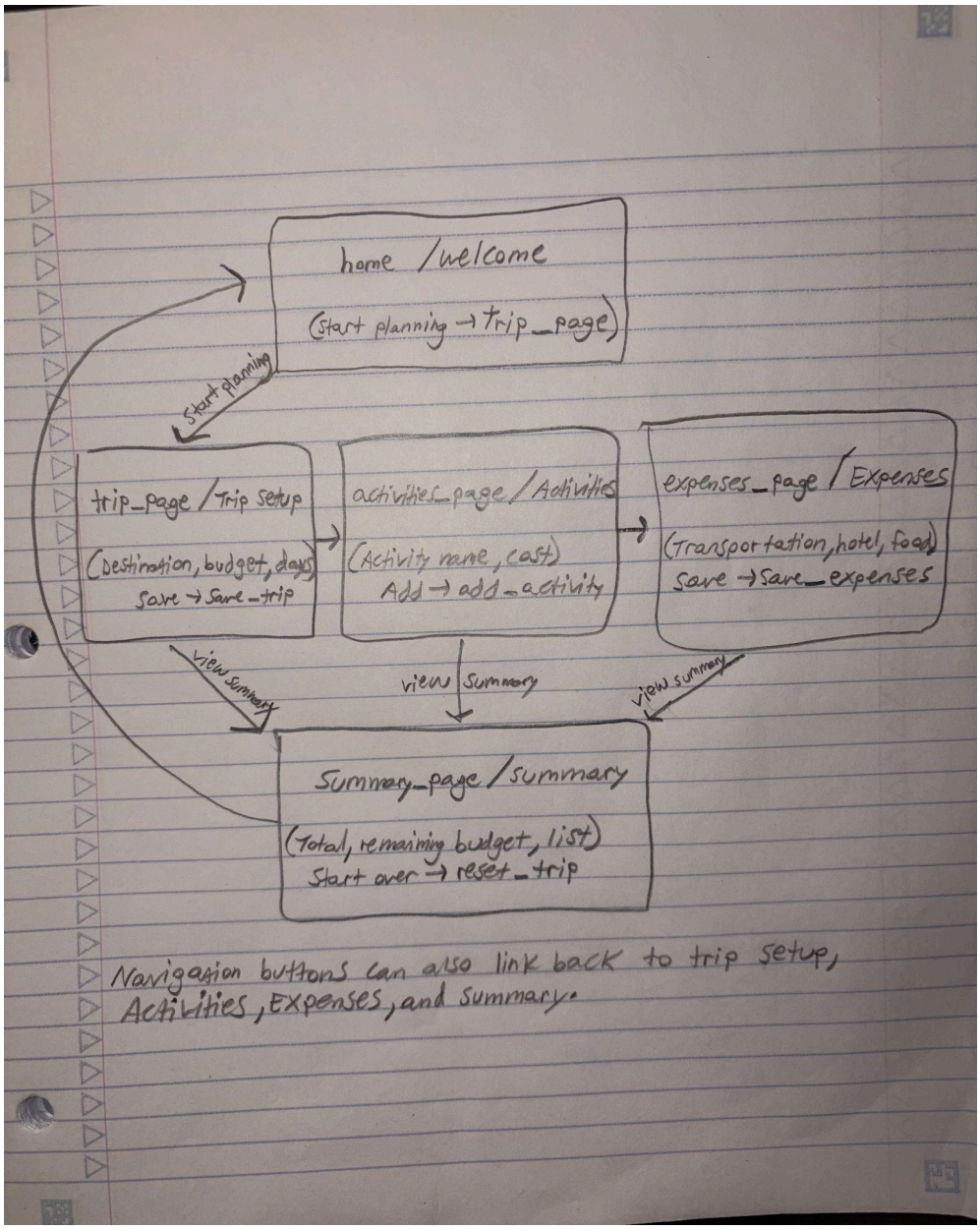
# 1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website. Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

**Description**: The rectangles are the five pages, and the arrows show the main buttons and routes. The names in each rectangle are my planned page functions.




## Planned State fields and types

destination: str; budget: float; days: int; activities: list[Activity]; transportation: float; accommodations: float; food: float. These values will stay in State as the user moves between the pages. Activity will have name: str and cost: float. The list of Activity instances will be updated when a user adds an activity.

Inputs to save_trip: destination, budget, days. Inputs to add_activity: activity name, activity cost. Inputs to save_expenses: transportation, accommodations, food. The page routes receive the current State.

# 2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

| **Requirement** | **Implementation** |
|---|---|
| **Purpose/Functionality**<br><br>A 1-2 sentence description of the purpose or functionality of your site | My website will help users, especially college students, plan a trip without going over their budget. They can enter basic trip information, add activities and estimated expenses, and see their total cost and how much money they have left. |
| **State Dataclass Attributes**<br><br>Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass.<br><br>Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site. | destination (str): changed when the user saves a destination.<br>budget (float): changed when the user saves or edits their budget.<br>days (int): changed when the user saves the number of travel days.<br>activities (list[Activity]): changed when the user adds an activity.<br>transportation (float), accommodations (float), and food (float): changed when the user saves expense estimates.<br>The reset_trip route will also clear these values for a new trip. These are real user choices, not constants. |
| **Second Dataclass**<br><br>Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified) | Activity will have a name (str) and estimated cost (float). When the user adds an activity, the website will create an Activity instance using their inputs and add it to the activities list in State. The names and costs will be different depending on what the user enters. |
| **Routes** (minimum 7)<br><br>The names of your route functions | 1. home – displays the welcome page.<br>2. trip_page – shows the trip setup page.<br>3. save_trip – saves destination, budget, and days.<br>4. activities_page – shows the activity list and inputs.<br>5. add_activity – adds an Activity to the list. |

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| | 6. expenses_page – shows the expense form.<br>7. save_expenses – updates transportation, accommodations, and food costs.<br>8. summary_page – calculates and displays the trip summary.<br>9. reset_trip – starts a new trip by clearing the saved information. |
| **Pages** (minimum 5)<br><br>A brief description of the pages in your site | 1. Welcome: explains the website and has a Start Planning button.<br>2. Trip Setup: asks for destination, budget, and number of days.<br>3. Activities: allows users to enter activity names and costs and see their list.<br>4. Expenses: allows users to enter transportation, accommodations, and food costs.<br>5. Trip Summary: shows the destination, activity list, total estimated cost, remaining budget, and a budget message. |
| **User Input** (minimum 3)<br><br>Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes<br><br>For more information on these input types, see Drafter Documentation. | TextBox for the destination and activity name.<br>TextBox for the trip budget, number of days, activity cost, and basic expense amounts; the program will convert numerical text to numbers.<br>SelectBox for an optional expense category if I decide to expand the expense form.<br>The required inputs are the TextBoxes above, so my basic site will still meet the requirement without the optional SelectBox. |
| **If Statements** (minimum 3)<br>Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site | 1. If the budget is not a valid positive amount, show a message asking the user to correct it instead of saving it.<br>2. If an activity name is blank or its cost is invalid or negative, do not add the activity and show a message.<br>3. If the estimated trip cost is greater than the budget, tell the user they are over budget; otherwise, show how much money they have left.<br>I can also check whether the number of travel days is positive. |

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| **Loop** (minimum 1)<br>This loop must iterate over the list/dict of dataclass instances in your State attribute.<br>Description of this loop and why it is meaningful to the site | I will use a for loop to go through the list of Activity instances saved in State. The loop will add each activity's cost to the activity total and help display the activity names and prices on the summary page. This is important because each user may add a different number of activities. |

# LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

*Below is my chat GPT conversation:*

***Student:*** *For my CISC106 final project, I am creating a travel budgeting website using Python and Drafter. I need at least three meaningful if statements and one loop. I want to come up with the ideas myself, but can you help me brainstorm some possible ways I could use these concepts in my website? I don't want any code yet.*

***ChatGPT:*** *Of course! Since your website focuses on tracking travel expenses and helping users stay within their budgets, you could use if statements to make decisions based on the information users enter.*

*For example, you could consider checking whether someone's expenses are greater than their budget, whether they have entered a valid budget, or whether they have added any activities to their trip.*

*For loops, think about information that users might enter multiple times. Since your website allows users to add activities, a loop could go through their activity list to calculate costs or display their plans.*

*Which of these ideas do you think would be most useful for your website?*

***Student:*** *I definitely want one of the if statements to check if the total cost of the trip is greater than the user's budget. If that is true, I want the website to notify them that they are over budget. I need at least two more if statements, and I don't want them to all do basically the same thing. What other situations could I check?*

***ChatGPT:*** *That makes sense. Your first if statement would help users understand their*

*spending.*

*For your second one, you could check whether the user enters a valid number for their budget. This would help prevent incorrect information from being used in calculations.*

*For your third one, you could check whether the user has added any activities. If they haven't, the website could display a message encouraging them to add activities before reviewing their trip.*

*These three conditions would have different purposes: comparing costs, checking input, and responding to the user's activity list.*

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***Student:*** *I think checking if the budget is greater than zero would be useful because someone shouldn't be able to enter a negative budget. I also like checking whether the activity list is empty. Would those count as meaningful if statements instead of just adding conditions to meet the requirement?*

***ChatGPT:*** *Yes, because each condition would affect how your website works.*

*The budget check prevents users from saving an invalid budget. The empty activity list check changes what the website displays when no activities have been entered. The spending comparison tells users whether their planned expenses fit within their budget.*

*They each solve a different problem, which makes them useful to your website rather than unnecessary conditions.*

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***Student:*** *Now I want to figure out the loop. Since users will be able to add multiple activities with different prices, I was thinking about using a loop to go through every activity and add the costs together. Would that work with the requirement about having a list of dataclass instances?*

***ChatGPT:*** *Yes. You could create an* `Activity` *dataclass that stores an activity's name and estimated cost. Your* `State` *data class could then have a list containing multiple* `Activity` *instances.*

*A loop could go through that list, access the estimated cost of each activity, and add those costs to a running total.*

*This would be meaningful because the total activity cost changes as users add more activities. It would also help meet the requirement that your loop uses a list of dataclass instances.*

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***Student:*** *I want users to see the activities they added, not just the total cost. Could I use the same list to show each activity's name and price on the website, or would I need to make another list?*

***ChatGPT:*** *You could use the same list because each* `Activity` *instance would already contain the name and cost.*

*You could use a loop to go through the activities and display that information. You wouldn't need a separate list containing the same details.*

*This would keep your project simpler and help you practice accessing attributes from dataclass instances.*

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***Student:*** *I think I want to use three if statements: checking if the user is over budget, checking if the budget is valid, and checking whether the activity list is empty. For my loop, I want to go through the list of activities and calculate their total cost. I also want to display the activity names and prices. Does this sound realistic?*

***ChatGPT:*** *Yes. These ideas use introductory Python concepts such as conditionals, loops, lists, dataclasses, and basic calculations.*

*They also connect directly to your website's purpose rather than adding unnecessary features. You can implement them using the Drafter library as you build your pages and routes.*

*One thing to remember is that the project also requires at least four fields in your* `State` *dataclass to be meaningfully modified. Your conditions and loop will help with the website's functionality, but you should plan your state fields and routes separately to make sure you meet those requirements.*

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***Student:*** *I'm going to stick with these ideas because I want my website to be useful but not too complicated. I would rather understand how everything works than add features that we haven't learned in class. I'll use these ideas when I start designing my routes and pages.*

***ChatGPT:*** *That's a reasonable approach. Focusing on a smaller number of useful features will make it easier to understand and test your program. As you begin developing the website, you can test each condition and loop separately to make sure it works before connecting everything together.*

### *Reflection*

*This brainstorming helped me figure out how I could use if statements and loops in my travel budgeting website without making the project too complicated. I decided to use three if statements to check whether a trip is over budget, whether the user entered a valid budget, and whether the activity list is empty. I also decided to use a loop to go through the activity list and calculate the total estimated cost. ChatGPT helped me think through different possibilities, but I chose the features that I thought would be most useful and realistic for my project. I did not use AI to generate any code during this discussion, but just to help me generate ideas and how I should apply them into what I want my code to produce.*

# Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.